

Progressive Semantic Residual Quantization for Multimodal-Joint Interest Modeling in Music Recommendation

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Abstract

In music recommendation systems, multimodal interests learning is pivotal, which allows the model to capture nuanced preferences, including textual elements such as lyrics and various musical attributes like different instruments and melodies. Recently, methods that incorporate multimodal content features through semantic IDs have achieved promising results. However, existing methods suffer from two critical limitations: 1) intra-modal semantic degradation, where residual-based quantization processes gradually decouple discrete IDs from original content semantics, leading to semantic drift; and 2) inter-modal modeling gaps, where traditional fusion strategies either overlook modal-specific details or fail to capture cross-modal correlations, hindering comprehensive user interest modeling. To address these challenges, we propose a novel multimodal recommendation framework with two stages. In the first stage, our Progressive Semantic Residual Quantization (PSRQ) method generates modal-specific and joint semantic IDs by explicitly preserving prefix semantic feature. In the second stage, to model multimodal interest of users, a Multi-Codebook Cross-Attention (MCCA) network is designed to enable the model to simultaneously capture modal-specific interests and perceive cross-modal correlations. Extensive experiments on multiple real-world datasets demonstrate that our framework outperforms state-of-the-art baselines. This framework has been deployed on one of China's largest music streaming platforms, and online A/B tests confirm significant

improvements in commercial metrics, underscoring its practical value for industrial-scale recommendation systems.

CCS Concepts

• Information systems → Retrieval models and ranking.

Keywords

Music Recommendation, Multimodal, Quantization, Semantic ID

ACM Reference Format:

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1 Introduction

In contemporary music streaming platforms, users exhibit varying preferences across different musical modalities, such as lyrics, instrumental and melodic. Even among different demographic groups, the emphasis on modal interests can differ significantly. For instance, Fiore et al. [1] found that adults focus more on lyrics, while children prioritize melody. Further research [2, 3] has indicated that different musical modalities can have distinct impacts on users' emotions. However, traditional recommendation models primarily rely on collaborative filtering [4], focus on modeling user behavioral preference [5], and lack the learning of multimodal interests.

In recent years, with the continuous advancement of multimodal feature extraction techniques, an increasing number of studies have applied multimodal information to fields such as short video and music recommendations [6–10]. These studies have shown that integrating multimodal information can significantly enhance the performance of recommendation systems by capturing the nuanced preferences of users. As researchers have delved deeper into multimodal recommendation, a central insight has emerged that semantic representation spaces of different modalities, including visual,

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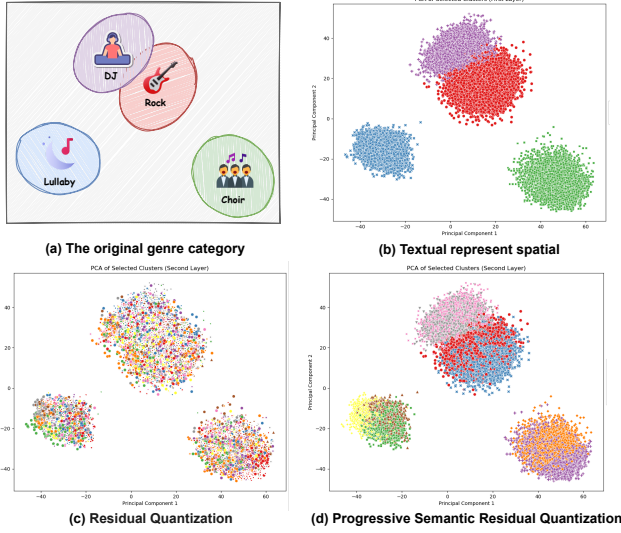


Figure 1: The subfigure 1(b) illustrates the original text feature space of songs in subfigure 1(a), visualized by t-SNE. Subfigure 1(c) displays the clustering results of the second layer in traditional residual quantization. Subfigure 1(d) presents the clustering results of the second layer using the proposed PSRQ method.

textual, and acoustic features, exhibit pronounced disparities[11–13]. These differences can hinder the integration of cross-modal cues, which are crucial for understanding the subtleties of user preferences[14–16]. In addition, traditional content representations are inherently static, as they are precomputed and fixed during training, posing a challenge as they cannot be optimized within an end-to-end recommendation framework. This limitation may impede the model’s adaptability to complex interaction patterns and could result in slower convergence during training[17].

Recently, quantization techniques have been widely applied in various fields and achieved remarkable results [18–20]. Among them, VQ-Rec [21] identified key challenges in vector quantization methods for recommender systems (VQ4Rec) and presents promising opportunities that can inspire future research in this emerging field. Tiger [22] further introduced the Residual Quantized Variational AutoEncoder (RQ-VAE[23]), officially opening the door of content feature transform into semantic IDs representation in the field of recommendation, through the application of the codebook[23, 24]. Subsequently, numerous research works[25–29] have shown that semantic IDs representation can significantly enhance the model’s ability to learn the representations of users’ multimodal interests, and alleviate the cold-start problem[30].

Despite these advancements, two critical challenges remain:

- **intra-semantic of multimodal:** Current approaches rely purely on geometric similarity (e.g., Euclidean distance or cosine similarity between residuals) for quantization. While Residual Quantization (RQ)[31] and RQ-VAE improve the accuracy of semantic IDs through iterative residual approximation. However, their layer-wise quantization process inherently decouples residual

vectors from original semantic meanings, overlooks hierarchical semantic alignment—the deeper the quantization layer, the weaker the connection to the original semantics. As shown in Figure 1 (c), the residual vectors can lead to more diverse and discrete clustering results, but tend to overlook the associations with the original semantics. Consequently, the generated cluster IDs may deviate from the intended item semantics, leading to suboptimal recommendation performance.

- **inter-semantic of multimodal:** Existing paradigms such as QARM[27] and OneRec[32] first fuse multimodal features via contrastive learning before quantization, which inevitably suppresses modal-unique signals critical for fine-grained user preference modeling. While M3CRS [29] preserves modal-specific characteristics through independent embedding table, its isolated modeling the modal-specific interest of user fails to capture cross-modal synergies (e.g., audio-lyrics complementarity in music). However, in the context of recommendation systems, both aspects are of paramount importance[14, 17, 29, 33, 34]. Therefore, the second challenge is how to simultaneously capture fine-grained modal preferences and exploit complementary cross-modal correlations for multimodal interest modeling based on semantic IDs after quantization.

To address these challenges, we propose a multimodal quantization-based recommendation framework that enhances both semantic fidelity and cross-modal interaction. In the feature engineering stage, we use a novel Progressive Semantic Residual Quantization (PSRQ) method preprocesses multimodal embeddings by explicitly preserving prefix semantic feature, generating modal-specific and joint semantic IDs that maintain strong alignment with original semantics. Then, for the multimodal interest modeling of user, we introduce the Multi-Codebook Cross-Attention(MCCA) Network, which employs a shared modal-joint codebook as a cross-modal query to model modal-specific embedding sequences. This design enables the model to simultaneously capture fine-grained modal preferences and exploit complementary cross-modal correlations. This approach operates end-to-end in the ranking stage[35] of the recommendation system, jointly optimizing for semantic consistency and adaptive multimodal fusion to achieve superior recommendation performance.

In summary, the contributions of our study are as follows:

- We proposed a novel Progressive Semantic Residual Quantization method that constrains residual quantization with prefix semantics, enhancing semantic preserve.
- We proposed a Multi-Codebook Cross-Attention Network for multimodal interest learning, simultaneously capturing modality specificity and cross-modal associations.
- Extensive offline experiments on three real-world datasets and online A/B tests verified the effectiveness of proposed method, significantly improving cold start performance metrics.

2 Related Work

2.1 Quantitative Representation Learning for Recommendation

Quantitative representation learning has recently attracted extensive attention from numerous scholars due to its ability to extract

semantic information, and its effectiveness has been demonstrated in multiple fields. VQ-Rec[21] converts text content vectors into sparse semantic ID representations through product quantization (PQ). TIGER[22] further utilizes RQ-VAE to generate hierarchical semantic IDs based on the text content feature as item representations. Although RQ and RQ-VAE approximate the original embeddings through residuals to improve the accuracy of quantitative representations, they still face the hourglass problem[36], resulting in uneven distribution in the discrete space and limited deep-level semantic associations. OneRec[32] enforces a uniform distribution of the number of elements in each layer of RQ through a multi-level balanced quantitative mechanism. Singh et al. [37] illustrated that Semantic IDs with Sentence Piece Models (SPM)[38] is a more adaptive and efficient solution for representing item content and achieve better generalization outcomes. Furthermore, Zheng et al. [39] propose a prefix ngram parameterization method and prove through experiments that incorporating the hierarchical nature of clustering into the embedding table mapping is an effective measure. Inspired by these works, we propose PSRQ and improve the association between residual layers and the original semantic information.

2.2 Multimodal Representation for Recommendation

In recent years, multimodal content features have achieved exciting results in enhancing recommendation systems. Early representative work, such as VBPR[6], introduced visual features into the recommendation field using matrix factorization. MMGCN[7] modeled the fine-grained modality of user preferences for each user-item bipartite graph of different modalities. To further improve multimodal based recommendation, the problem of multimodal interest representation fusion is crucial. A common approach is to fuse the multimodal embeddings of items through pre-training tasks. AlignRec[14] pre-trained the visual-text alignment task using a mask-then-predict strategy. Sheng et al. [17] refined users' multimodal interest representations through SimTier for the multimodal representations after contrastive learning pre-training tasks, and used MAKE to address the issue of the difference in training epochs required for ID features versus multimodal representations.

3 Preliminary

Problem Definition. Let $\mathcal{U}$ and $\mathcal{I}$ denote the sets of users and items, respectively. And $|\mathcal{U}|$ and $|\mathcal{I}|$ denote user and item number. For all items, we obtain their multimodal content embeddings $X^m \in \mathbb{R}^{|\mathcal{I}| \times d}$ based on existing content feature extraction methods. Here, $m \in \{v, t, a\}$, which represent visual, textual, and audio features, respectively. The specific content feature extraction methods are detailed in Section 5.1.1. For each user $u \in \mathcal{U}$, we construct his historical behavior sequences $\mathcal{H}_u = \{i_1^h, i_2^h, \dots, i_n^h\}$ based on positive interactions such as click, comment, or collect. In this sequence, $i_n^h \in \mathcal{I}$ denotes the item associated with the n -th interaction. Our recommendation task involves predicting the probability $\hat{y}_{u,t}$ that the user u will positively interact with the target item $i_t \in \mathcal{I}$.

Residual Quantization. In the conventional Kmeans-based Residual Quantization (RQ) process, each layer takes the residual vector from the previous layer as input and applies the Kmeans algorithm to obtain the cluster centers, which form the codebook

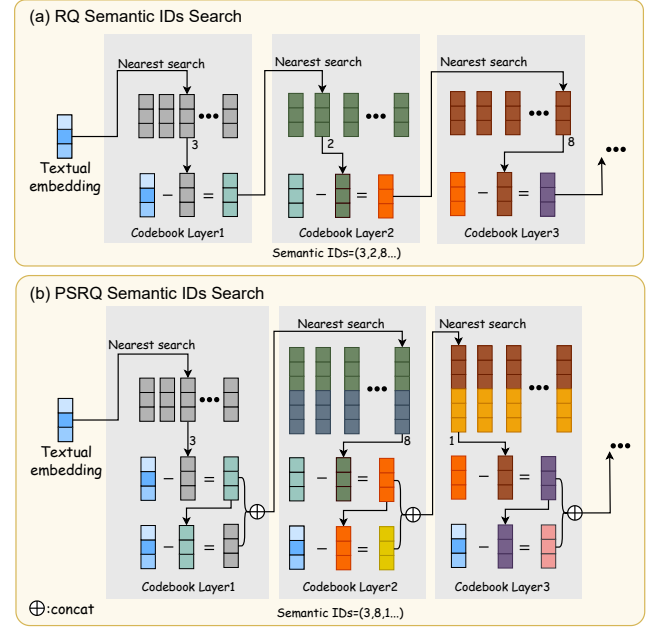


Figure 2: Subfigure 2(a) and 2(b) provide a visual comparison of the semantic ID retrieval processes in the RQ and PSRQ codebooks. It is assumed that the cluster number k of each layer is 8.

for the layer. For each kind of multimodal embeddings X_l^m :

$$\begin{aligned} C_1 &= \text{Kmeans}(X_1^m, k), X_1^m = X^m \\ C_2 &= \text{Kmeans}(X_2^m, k), X_2^m = X_1^m - \text{NearestRep}(X_1^m, C_1) \\ &\dots \\ C_l &= \text{Kmeans}(X_l^m, k), X_l^m = X_{l-1}^m - \text{NearestRep}(X_{l-1}^m, C_{l-1}) \end{aligned} \quad (1)$$

where l is the number of quantization layers, $C_l \in \mathbb{R}^{k \times d}$ are the generated cluster center embeddings of layer l , k is the number of cluster centers of Kmeans, and the $\text{NearestRep}(\cdot)$ denote the nearest representation search in the cluster center embeddings. The semantic IDs retrieval process of RQ is shown in Figure 2 (a).

4 Methodology

In this section, we elaborate on the components of our framework and its overall deployment workflow, depicted in Figure 3, which consists of two stages, feature engineering and downstream recommendation model training. During the feature engineering stage, multimodal raw data undergo multimodal encoder and then through PSRQ to obtain multi-codebooks. These codebooks generated semantic IDs representations and then fed into the downstream recommendation model, which is tasked with modeling user interests and predicting whether there will be a positive interaction with the target item.

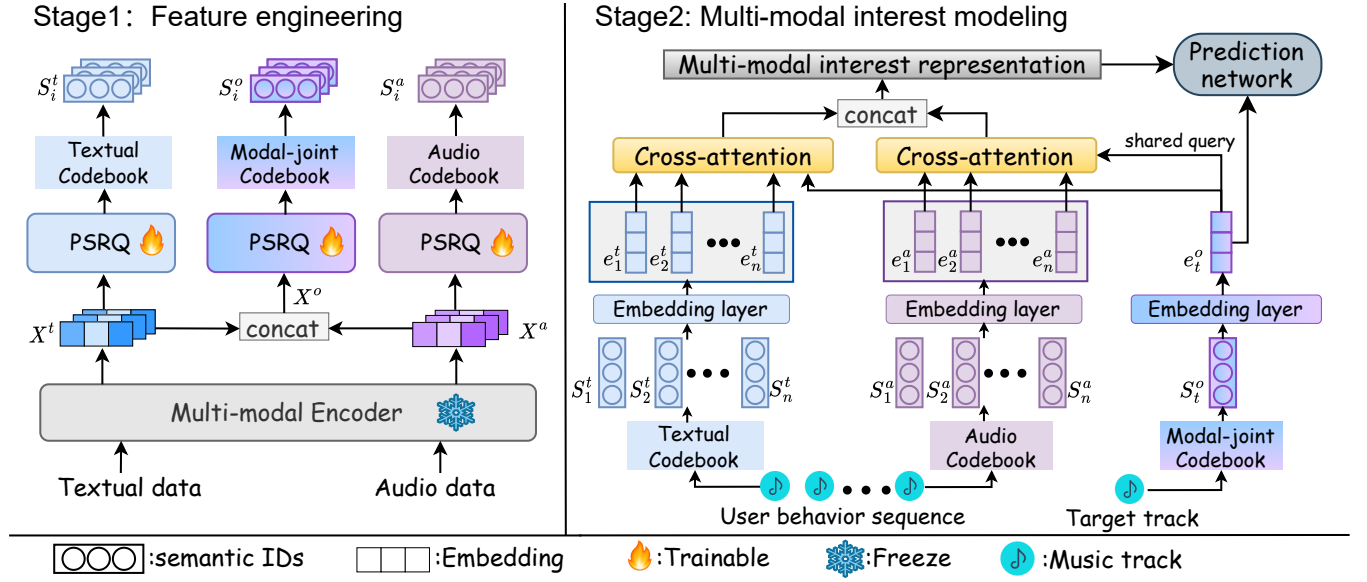


Figure 3: The overall workflow of our recommendation framework.

4.1 Progressive Semantic Residual Quantization

In the feature engineering phase, inspired by previous work [27, 29], we do not directly utilize original static content multimodal content embeddings $X^m \in \mathbb{R}^{|I| \times d}$. Instead, we employ our proposed Progressive Semantic Residual Quantization (PSRQ) method to map these embeddings to semantic ID representations. Different with RQ, the PSRQ introduces a critical modification that the residual vector is differentiated from the original content feature vector and then concatenated with it for enhancing the retention of the original semantic information, as detailed in the following equations:

$$\begin{aligned}
 X_1^m &= X^m \\
 C_1 &= \text{Kmeans}(X_1^m, k) \\
 X_2^m &= X^m - \text{NearestRep}(X_1^m, C_1) \\
 C_2 &= \text{Kmeans}(X_2^m \oplus (X^m - X_2^m), k) \\
 &\dots \\
 X_l^m &= X_{l-1}^m - \text{NearestRep}(X_{l-1}^m, C_{l-1}) \\
 C_l &= \text{Kmeans}(X_l^m \oplus (X^m - X_l^m), k)
 \end{aligned} \tag{2}$$

where $\oplus$ denote concat operation and $C^1 \in \mathbb{R}^{k \times d}$, C^2 to C^l all $\in \mathbb{R}^{k \times 2d}$.

Then for each item i , we retrieve the nearest cluster ID $id^t \in (0, 1, \dots, k-1)$ from each quantization layer as the semantic IDs $S_i^m = [id_1^m, id_2^m, \dots, id_l^m]$. The different of semantic IDs retrieve process between conventional RQ and PSRQ is shown in Figure 2.

In addition, we concatenate textual and audio modal embeddings as modal-joint embeddings $X^o \in \mathbb{R}^{|I| \times 2d}$ to perform PSRQ and generate modal-joint semantic IDs S_i^o for each item.

4.2 Multi-Codebook Cross-Attention

Following the quantization of modal-specific and joint content representations, we integrate semantic IDs into collaborative filtering-based recommendation models. This integration enables the modeling of user multimodal interests, enhancing the generalization capability.

4.2.1 Hierarchical embedding layer. To enable end-to-end optimization of content features beyond the constraints of static representations, we do not use the original cluster centroid embeddings as semantic IDs embedding. Instead, within the embedding layer of our model, we use randomly initialized embedding tables for both the modal-specific and joint semantic IDs generated through the quantization codebooks. Specifically, for the modal-specific semantic IDs S_i^m of each item i^h in the user behavior sequence $\mathcal{H}_u$, we allocate a randomly initialized embedding table denoted as $E^m \in \mathbb{R}^{k \times d'}$ for each quantization layer, where d' is the embedding size. Then we can retrieve the semantic ID embeddings of each layer and aggregate them as the final semantic representation $e_i^m \in \mathbb{R}^{d'}$ of each item.

$$e_i^m = \sum_{j=1}^l \text{one-hot}(id_{ij}^m) \times E_j^m, id_{ij}^m \in S_i^m \tag{3}$$

where one-hot is a commonly used option, which encode the id_{ij} into a one-hot vector. As this method, we form the semantic embedding sequence $\{e_1^m, e_2^m, \dots, e_n^m\}$ of the user.

For the target item i_t , we use the modal-joint codebook to obtain its semantic embedding $e_i^o \in \mathbb{R}^{d'}$, which aims to capture the cross-modal associations.

Additionally, we integrate collaborative signals by leveraging the sequence of ID embeddings $\{e_1^{\text{col}}, e_2^{\text{col}}, \dots, e_n^{\text{col}}\}$ and the target item ID embedding $e_i^{\text{col}} \in \mathbb{R}^{d'}$. This approach ensures that the model effectively captures the collaborative patterns within the data.

4.2.2 Cross-attention layer. After obtaining the user modal-specific embedding sequences and the modal-joint embedding e_t^o of the target item, we use the cross-attention mechanism to model the user modal-specific interests while capturing cross-modal correlations. The modal-joint semantic IDs embedding of the target item is utilized as a shared query to calculate attention scores over the modal-specific embedding sequences of the user. This design is capable of alleviating the issue of inconsistent representation spaces across modalities to a certain extent, while also supporting the capture of cross-modal correlations. The multimodal interest representation $h_u^m \in \mathbb{R}^{d'}$ are then computed via attention-weighted aggregation:

$$\begin{aligned} h_u^m &= \sum_{j=1}^n \text{Cross-Attention}(Q = e_t^o, K = e_j^m, v = e_j^m) \\ &= \sum_{j=1}^n \text{Attention}(e_j^m \oplus e_t^o) \cdot e_j^m \end{aligned} \quad (4)$$

where $\text{Attention}(\cdot)$ is a feed-forward network with output as the activation weight.

To capture the collaborative patterns, we also apply the attention mechanism to the collaborative embedding sequence $\mathcal{H}_u^{\text{col}}$ and then obtain the collaborative interest representation of the user:

$$h_u^{\text{col}} = \sum_{j=1}^n \text{Attention}(e_j^{\text{col}} \oplus e_t^{\text{col}}) \cdot e_j^{\text{col}} \quad (5)$$

where h_u^{col} is collaborative interest representation of the user.

4.3 Model Prediction & Optimization

To derive the probability of positive interaction between the user and the target item, we concatenate the multimodal interest representations h_u^m , collaborative interest representation h_u^{col} with the target item's collaborative embedding e_t^{col} and modal-joint semantic IDs embedding e_t^o . This concatenated vector is then fed into a Multi-layer Perceptron (MLP) for the predicted logit $\hat{y}_j$. Since the positive interaction prediction is a binary classification task, we employ cross-entropy loss as the objective function for model training and optimization:

$$\mathcal{L} = -\frac{1}{N} \sum_{j=1}^N y_j \log \sigma(\hat{y}_j) + (1 - y_j) \log(1 - \sigma(\hat{y}_j)) \quad (6)$$

where N is the total number of training instances, and $y_j \in \{0, 1\}$ is the label for each sample.

5 Experiments

In this section, we conduct a variety of offline experiments and online tests to evaluate the proposed method. Specifically, our aim is to address the following research questions.

- **RQ1** How does the proposed quantization-based recommendation framework compare in performance to general and SOTA multimodal recommendation methods?
- **RQ2** How does the semantic IDs generated from PSRQ method perform in recommendation tasks compared to other quantization methods?
- **RQ3** Does the modal-specific and modal-joint codebooks in MCCA effectiveness

- **RQ4** How does the overall performance of the proposed method (PSRQ+MCCA) fare in real-world online scenarios?

5.1 Experimental Settings

Table 1: Statistics of datasets

Dataset	#Users	#Items	#Interactions
Amazon baby	81,423	33,652	230,444
Industrial	4,926,656	1,387,247	8,696,093
Music4all	14,127	99,596	2,597,382

5.1.1 Datasets. To evaluate the performance and effectiveness of the proposed method, we conduct experiments on an industrial dataset and two public datasets, including Amazon Baby[40], and Music4all[41]. The detailed data statistics and multimodal information of each dataset are presented in Table 1. For all dataset, we mark the items with less than 30 interaction times as cold items.

- **Amazon Baby:** For the baby benchmark in the Amazon review dataset, we define interactions with ratings of 4 or higher as positive samples and those with ratings below 4 as negative samples. User interaction sequences are constructed based on timestamps. The image features of items are extracted using the pre-existing CNN-based method provided by the dataset. Text features are obtained by the LLaMA3.2-1B[42] model from the text composed of the product title, description, brand, and categorical information.
- **Industrial:** This dataset is collected from our music platform over a one-week period. We consider user-song collection interactions as positive samples and non-collected tracks as negative samples, constructing user historical interaction sequences from positive interaction. Textual embeddings are extracted by the Baichuan2-7B[43] model from textual information including titles, genre tags, and lyrics. The audio embeddings are extracted from MP3 files using the MERT-v1-95M[44] model.
- **Music4all:** We define repeatedly played songs as positive samples and songs played only once as negative samples. Target items with less than 30 interactions are regarded cold. Text features are extracted by LLaMA3.2 from song titles, lyrics, genre tags, etc. The audio features are also sampled and extracted by the MERT model.

5.1.2 Implementation Details. We implement all methods in TensorFlow 2, while the epoch number is set to one, the training batch sizes in three datasets are $\{64, 512, 512\}$ and the learning rates are $\{0.0005, 0.0001, 0.0001\}$. The dimension d' of both the multimodal semantic IDs embedding and ID embedding is set to 64. The maximum length of the user history sequence is truncated to 20. For all models, we adopt the Adam [45] optimizer. All of the Multimodal Large Language Models[46] (MLLMs), including Baichuan2-7B, MERT-v1-95M, and LLaMA3.2-1B, are deployed on NVIDIA A100 GPUs for extracting multimodal content embeddings. For the proposed PSRQ and other quantization methods, we set the number of clusters k in Amazon Baby, Industrial, and Music4all at $\{64, 256, 128\}$, and the residual layer number l of RQ, PQ, RQ-VAE, and PSRQ are $\{3, 4, 3, 3\}$.

Table 2: Performance across baselines on various datasets is evaluated by AUC metrics for all and cold-start items, with model fit indicated by Logloss. The top performer is bolded, and the second by underlines. "Improv." indicates relative improvements over the best baseline in percentage.

Methods	Amazon Baby			Industrial			Music4all		
	All AUC	Cold AUC	Logloss	All AUC	Cold AUC	Logloss	All AUC	Cold AUC	Logloss
VBPR	0.6466	0.5377	2.6145	0.7407	0.7229	1.4869	0.6217	0.5174	5.7460
SimTier+MAKE	0.6213	0.5286	2.7974	0.7537	<u>0.7446</u>	1.2503	0.6871	0.6041	3.6962
DIN	0.6492	0.5487	2.5766	0.7599	0.7382	1.2188	0.7260	0.6699	2.9386
QARM	0.6557	<u>0.5681</u>	2.5686	0.7628	0.7429	1.2068	0.7347	0.7336	2.9181
PSRQ+MCCA	0.6573	0.5781	2.5564	0.7636	0.7535	1.2006	0.7347	0.7373	2.9070
%Improv.	+0.24%	+1.76%	-0.47%	+1.04%	+1.20%	-0.51%	+0.00%	+0.50%	-0.38%

Table 3: Augment only textual semantic IDs to the DIN model from various quantization methods to fairly assess the performance of each quantization approach. The top performer is bolded, and the second by underlines.

Methods	Amazon Baby			Industrial			Music4all		
	All AUC	Cold AUC	Logloss	All AUC	Cold AUC	Logloss	All AUC	Cold AUC	Logloss
DIN	0.6492	0.5487	2.5766	0.7599	0.7382	1.2188	0.7260	0.6699	2.9386
+PQ	0.6534	0.5515	2.5754	0.7617	0.7387	<u>1.2050</u>	0.7313	0.7321	2.9234
+VQ	0.6520	<u>0.5569</u>	2.5725	0.7623	<u>0.7411</u>	1.2087	0.7328	0.7317	2.9146
+RQ	0.6531	0.5546	2.5661	<u>0.7628</u>	0.7334	1.2077	0.7329	<u>0.7327</u>	2.9204
+RQ-VAE	<u>0.6535</u>	0.5533	2.5711	0.7620	0.7400	1.2081	<u>0.7338</u>	0.7322	2.9184
+PSRQ	0.6540	0.5610	<u>2.5705</u>	0.7630	0.7442	1.2003	0.7345	0.7331	<u>2.9183</u>

Furthermore, to ensure the fairness of the training process, we maintained consistent parameter scales, learning rates, and batch sizes across all models at different datasets.

5.1.3 Evaluation Metrics. To comprehensively evaluate the performance of models, we adopt the AUC (Area Under the Curve) [47] as the primary evaluation metric. In addition, under the same model parameter magnitude, we provide the Logloss metric, with the specific formula detailed in Section 4.3.

5.2 Offline Performance Comparison

5.2.1 Comparison with the recommendation models proposed for industry scenarios(RQ1). To validate the effectiveness of the proposed framework, we compare the performance of our model with various baselines, including the latest SOTA models.

- **DIN[5]:** This model utilizes attention mechanism to dynamically capture users' interests from their historical behaviors.
- **VBPR[6]:** This model integrates the multimodal embeddings and ID embeddings of each item as its representation and uses the matrix factorization (MF) framework to reconstruct the historical interactions between users and items.
- **SimTier+MAKE[17]:** An industrial-grade recommendation framework that combines similarity tiering and multimodal knowledge embedding to handle large-scale heterogeneous data.
- **QARM[27]:** A recent SOTA approach that integrates contrastive learning and quantization techniques to enhance recommendation efficiency while preserving model expressiveness.

As shown in Table 2, the proposed PSRQ+MCCA model achieves superior performance across most experimental results on the three datasets, particularly excelling in the recommendation of cold-start items. QARM, which integrates contrastive learning and quantization techniques to enhance semantic generalization, attains the second-best performance across multiple datasets. DIN, relying solely on user behavior interest for prediction, demonstrates good fitting due to the thorough end-to-end training of ID embeddings, despite the lack of multimodal information. In contrast, SimTier+MAKE, despite incorporating an auxiliary network for learning multimodal embeddings, underperforms DIN (only outperforming VBPR) with relatively high Logloss, likely due to insufficient training within a single epoch.

5.2.2 Comparison of quantization method (RQ2). To rigorously assess the enhancement of the proposed PSRQ method over alternative quantization techniques in terms of content feature generalization and its impact on recommendation performance, we maintain identical batch sizes and learning rates alongside the DIN [5] model, with only textual modal embedding. This approach serves to mitigate the influence of extraneous factors, ensuring a focused evaluation of the quantization methods themselves. The quantization methods for comparison are as follows:

- **PQ[48]:** It is a widely used technique that compresses high-dimensional vectors into a lower-dimensional space by dividing the vector into subvectors and quantizing each subvector independently.

Table 4: Ablate the role of modality-specific and cross-modal shared queries within the MCCA framework.

Methods	Amazon Baby			Industrial			Music4all		
	All AUC	Cold AUC	Logloss	All AUC	Cold AUC	Logloss	All AUC	Cold AUC	Logloss
w/o MSC	0.6544	0.5549	2.5594	0.7627	0.7424	1.1991	0.7333	0.7281	2.9184
w/o MJC	<u>0.6555</u>	<u>0.5619</u>	2.5598	<u>0.7631</u>	<u>0.7467</u>	<u>1.2003</u>	0.7353	<u>0.7364</u>	<u>2.9089</u>
PSRQ+MCCA	0.6573	0.5781	2.5564	0.7636	0.7535	1.2006	<u>0.7347</u>	0.7373	2.9070

- **VQ[49]**: This method involves compressing high-dimensional vectors into a lower-dimensional space using a codebook generated by Kmeans clustering.
- **RQ[50]**: Similarly to VQ, this RQ is also based on Kmeans clustering but focuses on quantization residuals iteratively to achieve more accurate approximations of the original vectors. This technique enhances the precision of vector quantization and is particularly effective in capturing fine-grained semantic differences between items.
- **RQ-VAE[23]**: This approach extends the concept of RQ by integrating it with an autoencoder architecture. RQ-VAE not only quantizes the vectors but also learns to reconstruct them, which helps in preserving more semantic information compared to traditional quantization methods.

Referring to Table 3, all quantization methods have enhanced the performance of the DIN model, benefiting from the end-to-end training of semantic ID embeddings. Among them, PQ, due to its lack of global semantic information, resulted in the poorest performance. VQ, despite employing only a single-layer quantization, achieved considerable improvements for cold-start items. Both RQ and RQ-VAE achieved the second-best results across multiple datasets. The PSRQ method demonstrated the overall best performance, including for all items and cold-start items in three datasets.

5.2.3 Ablation Study of MCCA (RQ3). To validate the effectiveness of MCCA, we conduct ablation studies, aiming to isolate the impact of modal-specific and joint:

- **w/o modal-specific codebooks(w/o MSC)**: We employ a shared codebook for all modalities during attention modeling, eliminating the dedicated modal-specific interest representations. This ablation tests whether removing granular modal semantics weakens the ability of modal to capture fine-grained user preferences across different content modalities.
- **w/o modal-joint codebooks(w/o MJC)**: To assess the significance of cross-modal correlation modeling, we conducted experiments by excluding the shared modal-joint query for each modal sequence. Instead, we utilized modal-specific semantic IDs embeddings to serve as queries for the respective modality's semantic embedding sequences. This variant investigates whether the absence of a shared query mechanism hinders the capacity to exploit inter-modal dependencies, and affecting recommendation performance.

The results of the ablation study, as shown in Table 4, indicate that the MCCA framework outperforms on most datasets and metrics, validating the effectiveness of extracting cross-modal information through modal-joint codebooks and modeling each modality

independently. Although on the Music4all dataset, MCCA's performance for all items is slightly inferior to that of w/o MJC, the improvements for cold-start items demonstrate that cross-modal associations are beneficial for enhancing the model's generalization performance, particularly for cold-start items.

5.3 Online A/B Tests (RQ4)

We conducted an A/B test on our music streaming platform's online ranking modal from Feb. 24 to Mar. 02, 2025, delivering song recommendations to tens of millions of users daily. The baseline model was the industry-standard Deep Learning Recommendation Model(DLRM[51]), which integrates heterogeneous features like user profiles, historical interactions, and collaborative filtering signals. The experiment group enhanced the baseline by incorporating user multimodal interest representations, derived from PSRQ generated semantic IDs and MCCA, into the user's context features. These were concatenated with other user features and fed into a multi-task prediction network. Our commercial metrics include user engagement probabilities with music tracks, specifically through behaviors such as *collect* and *full_play*. During the trial period, the experiment group saw a 0.7% increase in *collect* and a 0.9% increase in *full_play* compared to the control group. For songs released within the last 30 days, the probabilities of *collect* and *full_play* increased by 1.4% and 2.2%, respectively.

6 Conclusion

In this work, we introduced a novel multimodal recommendation framework to address the persistent challenges of semantic degradation and cross-modal modeling gaps in music recommendation systems. Our Progressive Semantic Residual Quantization (PSRQ) method effectively preserves original semantic during quantization, while the Multi-Codebook Cross-Attention (MCCA) mechanism enables simultaneous capture of fine-grained modal-specific interests and cross-modal correlations. Extensive experiments on industrial and public datasets demonstrated significant improvements in recommendation accuracy and cold start robustness, validating state-of-the-art performance of our framework. The successful deployment on a leading music streaming platform underscores its practical value in real-world scenarios. Future work could explore dynamic codebook adaptation and broader applications in multimodal content recommendation. This study advances the field by bridging semantic fidelity and multimodal synergy, offering a scalable solution for industrial recommendation systems.

References

- [1] Jennifer Fiore. Analysis of lyrics from group songwriting with bereaved children and adolescents. *Journal of Music Therapy*, 53(3):207–231, 05 2016.

- [2] Yiqing Ma, David Baker, Katherine Vukovics, Connor Davis, and Emily Elliott. Generalizing the effect of lyrics on emotion rating. 07 2021.
- [3] Soravitt Sangnark, Phairot Authasan, Puntawat Ponglertnapakorn, Phudit Chalekarn, Thapanun Sudhawiyangkul, Manatsanan Trakulruangroj, Sarita Songsermsawad, Rawin Assabumrungrat, Supalak Ampod, Kajornvut Ounjai, and Theerawit Wilaiprasitporn. Revealing preference in popular music through familiarity and brain response. *IEEE Sensors Journal*, 21(13):14931–14940, 2021.
- [4] G. Linden, B. Smith, and J. York. Amazon.com recommendations: item-to-item collaborative filtering. *IEEE Internet Computing*, 7(1):76–80, 2003.
- [5] Guorui Zhou, Xiaoqiang Zhu, Chenru Song, Ying Fan, Han Zhu, Xiao Ma, Yanghui Yan, Junqi Jin, Han Li, and Kun Gai. Deep interest network for click-through rate prediction. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, KDD '18*, page 1059–1068, New York, NY, USA, 2018. Association for Computing Machinery.
- [6] Ruining He and Julian McAuley. Vbpr: visual bayesian personalized ranking from implicit feedback. In *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence, AAAI'16*, page 144–150. AAAI Press, 2016.
- [7] Yinwei Wei, Xiang Wang, Liqiang Nie, Xiangnan He, Richang Hong, and Tat-Seng Chua. Mmgn: Multi-modal graph convolution network for personalized recommendation of micro-video. In *Proceedings of the 27th ACM International Conference on Multimedia, MM '19*, page 1437–1445, New York, NY, USA, 2019. Association for Computing Machinery.
- [8] Jiashuo Yu, Jinyu Liu, Ying Cheng, Rui Feng, and Yuejie Zhang. Modality-aware contrastive instance learning with self-distillation for weakly-supervised audio-visual violence detection. In *Proceedings of the 30th ACM International Conference on Multimedia, MM '22*, page 6278–6287, New York, NY, USA, 2022. Association for Computing Machinery.
- [9] Songpei Xu, Shijia Wang, Da Guo, Xianwen Guo, Qiang Xiao, Fangjian Li, and Chuanjiang Luo. An efficient large recommendation model: Towards a resource-optimal scaling law, 2025.
- [10] Qimeng Yang, Shijia Wang, Da Guo, Dongjin Yu, Qiang Xiao, Dongjing Wang, and Chuanjiang Luo. Cascading multimodal feature enhanced contrast learning for music recommendation. In *2024 IEEE International Conference on Data Mining (ICDM)*, pages 905–910, 2024.
- [11] Rohit Girdhar, Alaaeldin El-Nouby, Zhuang Liu, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, and Ishan Misra. Imagebind one embedding space to bind them all. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15180–15190, 2023.
- [12] Junnan Li, Ramprasaath R. Selvaraju, Akhilesh D. Gotmare, Shafiq Joty, Caiming Xiong, and Steven C.H. Hoi. Align before fuse: vision and language representation learning with momentum distillation. In *Proceedings of the 35th International Conference on Neural Information Processing Systems, NIPS '21*, Red Hook, NY, USA, 2021. Curran Associates Inc.
- [13] Ji Lin, Hongxu Yin, Wei Ping, Yao Lu, Pavlo Molchanov, Andrew Tao, Huizi Mao, Jan Kautz, Mohammad Shoeybi, and Song Han. Vila: On pre-training for visual language models, 2023.
- [14] Yifan Liu, Kangning Zhang, Xiangyuan Ren, Yanhua Huang, Jiarui Jin, Yingjie Qin, Ruilong Su, Ruiwen Xu, Yong Yu, and Weinan Zhang. Alignrec: Aligning and training in multimodal recommendations. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, CIKM '24*, page 1503–1512, New York, NY, USA, 2024. Association for Computing Machinery.
- [15] Hengchang Hu, Wei Guo, Yong Liu, and Min-Yen Kan. Adaptive multi-modalities fusion in sequential recommendation systems. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management, CIKM '23*, page 843–853, New York, NY, USA, 2023. Association for Computing Machinery.
- [16] D. Rafailidis, P. Kefalas, and Y. Manolopoulos. Preference dynamics with multi-modal user-item interactions in social media recommendation. *Expert Systems with Applications*, 74:11–18, 2017.
- [17] Xiang-Rong Sheng, Feifan Yang, Litong Gong, Biao Wang, Zhangming Chan, Yujing Zhang, Yueyao Cheng, Yong-Nan Zhu, Tiezheng Ge, Han Zhu, Yuning Jiang, Jian Xu, and Bo Zheng. Enhancing taobao display advertising with multi-modal representations: Challenges, approaches and insights. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, CIKM '24*, page 4858–4865, New York, NY, USA, 2024. Association for Computing Machinery.
- [18] Artem Babenko and Victor Lempitsky. Additive quantization for extreme vector compression. In *2014 IEEE Conference on Computer Vision and Pattern Recognition*, pages 931–938, 2014.
- [19] Yue Li, Wenrui Ding, Chunlei Liu, Baochang Zhang, and Guodong Guo. TRQ: Ternary neural networks with residual quantization. In *Proceedings of the Thirty-Fifth AAAI Conference on Artificial Intelligence*, volume 35, pages 8538–8546, Palo Alto, California, USA, 2021. AAAI Press. Accessed: 2025-05-05.
- [20] Zefan Li, Bingbing Ni, Wenjun Zhang, Xiaokang Yang, and Wen Gao. Performance guaranteed network acceleration via high-order residual quantization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, Oct 2017.
- [21] Yupeng Hou, Zhankui He, Julian McAuley, and Wayne Xin Zhao. Learning vector-quantized item representation for transferable sequential recommenders. In *Proceedings of the ACM Web Conference 2023, WWW '23*, page 1162–1171, New York, NY, USA, 2023. Association for Computing Machinery.
- [22] Shashank Rajput, Nikhil Mehta, Anima Singh, Raghunandan Keshavan, Trung Vu, Lukasz Heldt, Lichan Hong, Yi Tay, Vinh Q. Tran, Jonah Samost, Maciej Kula, Ed H. Chi, and Maheswaran Sathiamoorthy. Recommender systems with generative retrieval. In *Proceedings of the 37th International Conference on Neural Information Processing Systems, NIPS '23*, Red Hook, NY, USA, 2023. Curran Associates Inc.
- [23] Doyup Lee, Chihyeon Kim, Saehoon Kim, Minsu Cho, and Wook-Shin Han. Autoregressive image generation using residual quantization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11523–11532, June 2022.
- [24] Julieta Martinez, Holger H. Hoos, and James J. Little. Stacked quantizers for compositional vector compression, 2014.
- [25] Yupeng Hou, Zhankui He, Julian McAuley, and Wayne Xin Zhao. Learning vector-quantized item representation for transferable sequential recommenders. In *Proceedings of the ACM Web Conference 2023, WWW '23*, page 1162–1171, New York, NY, USA, 2023. Association for Computing Machinery.
- [26] Anima Singh, Trung Vu, Nikhil Mehta, Raghunandan Keshavan, Maheswaran Sathiamoorthy, Yilin Zheng, Lichan Hong, Lukasz Heldt, Li Wei, Devansh Tandon, Ed Chi, and Xinyang Yi. Better generalization with semantic ids: A case study in ranking for recommendations. In *Proceedings of the 18th ACM Conference on Recommender Systems, RecSys '24*, page 1039–1044, New York, NY, USA, 2024. Association for Computing Machinery.
- [27] Xinchun Luo, Jiangxia Cao, Tianyu Sun, Jinkai Yu, Rui Huang, Wei Yuan, Hezheng Lin, Yichen Zheng, Shiyao Wang, Qigen Hu, Changqing Qiu, Jiaqi Zhang, Xu Zhang, Zhiheng Yan, Jingming Zhang, Simin Zhang, Mingxing Wen, Zhaojie Liu, Kun Gai, and Guorui Zhou. Qarm: Quantitative alignment multi-modal recommendation at kuaishou, 2024.
- [28] Guanghan Li, Xun Zhang, Yufei Zhang, Yifan Yin, Guojun Yin, and Wei Lin. Semantic convergence: Harmonizing recommender systems via two-stage alignment and behavioral semantic tokenization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 12040–12048, AAAI, 2025. AAAI Press.
- [29] Gaode Chen, Ruina Sun, Yuezhan Jiang, Jiangxia Cao, Qi Zhang, Jingjian Lin, Han Li, Kun Gai, and Xinghua Zhang. A multi-modal modeling framework for cold-start short-video recommendation. In *Proceedings of the 18th ACM Conference on Recommender Systems, RecSys '24*, page 391–400, New York, NY, USA, 2024. Association for Computing Machinery.
- [30] Zhihui Zhou, Lilin Zhang, and Ning Yang. Contrastive collaborative filtering for cold-start item recommendation. In *Proceedings of the ACM Web Conference 2023, WWW '23*, page 928–937, New York, NY, USA, 2023. Association for Computing Machinery.
- [31] Doyup Lee, Chihyeon Kim, Saehoon Kim, Minsu Cho, and Wook-Shin Han. Autoregressive image generation using residual quantization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11523–11532, 2022.
- [32] Jiaxin Deng, Shiyao Wang, Kuo Cai, Lejian Ren, Qigen Hu, Weifeng Ding, Qiang Luo, and Guorui Zhou. Onerec: Unifying retrieve and rank with generative recommender and iterative preference alignment, 2025.
- [33] Feiyu Chen, Junjie Wang, Yinwei Wei, Hai-Tao Zheng, and Jie Shao. Breaking isolation: Multimodal graph fusion for multimedia recommendation by edge-wise modulation. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 385–394, 2022.
- [34] Hang Zhou, Yucheng Wang, and Huijing Zhan. Mde: Modality discrimination enhancement for multi-modal recommendation, 2025.
- [35] Shijia Wang, Yi Zheng, Qiang Xiao, Yilong Zhao, Qimeng Yang, and Chuanjiang Luo. Sparsity-aware personalized pattern extractor network for music multi-task learning. In Makoto Onizuka, Jae-Gil Lee, Yongxin Tong, Chuan Xiao, Yoshiharu Ishikawa, Sihem Amer-Yahia, H. V. Jagadish, and Kejing Lu, editors, *Database Systems for Advanced Applications*, pages 352–363, Singapore, 2024. Springer Nature Singapore.
- [36] Zhirui Kuai, Zuxu Chen, Huimu Wang, Mingming Li, Dadong Miao, Binbin Wang, Xusong Chen, Li Kuang, Yuxing Han, Jiaxing Wang, Guoyu Tang, Lin Liu, Songlin Wang, and Jingwei Zhuo. Breaking the hourglass phenomenon of residual quantization: Enhancing the upper bound of generative retrieval. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 677–685, Miami, Florida, USA, 2024. Association for Computational Linguistics.
- [37] Anima Singh, Trung Vu, Nikhil Mehta, Raghunandan Keshavan, Maheswaran Sathiamoorthy, Yilin Zheng, Lichan Hong, Lukasz Heldt, Li Wei, Devansh Tandon, Ed Chi, and Xinyang Yi. Better generalization with semantic ids: A case study in ranking for recommendations. In *Proceedings of the 18th ACM Conference on Recommender Systems, RecSys '24*, page 1039–1044, New York, NY, USA, 2024. Association for Computing Machinery.
- [38] Taku Kudo. Subword regularization: Improving neural network translation models with multiple subword candidates. *arXiv preprint arXiv:1804.10959*, 2018.
- [39] Carolina Zheng, Minhui Huang, Dmitrii Pedchenko, Kaushik Rangadurai, Siyu Wang, Gaby Nahum, Jie Lei, Yang Yang, Tao Liu, Zutian Luo, et al. Enhancing

- embedding representation stability in recommendation systems with semantic id. *arXiv preprint arXiv:2504.02137*, 2025.
- [40] Yupeng Hou, Jiacheng Li, Zhankui He, An Yan, Xiusi Chen, and Julian McAuley. Bridging language and items for retrieval and recommendation, 2024.
- [41] Igor André Pegoraro Santana, Fabio Pinhelli, Juliano Donini, Leonardo Gabiato Catharin, Rafael B. Mangolin, Yandre M. G. Costa, Valéria Delisandra Feltrim, and Marcos Aurélio Domingues. Music4all: A new music database and its applications. *2020 International Conference on Systems, Signals and Image Processing (IWSSIP)*, pages 399–404, 2020.
- [42] Vladimir Malinovskii, Andrei Panferov, Ivan Ilin, Han Guo, Peter Richtárik, and Dan Alistarh. Pushing the limits of large language model quantization via the linearity theorem, 2024.
- [43] Aiyuan Yang, Bin Xiao, Bingning Wang, Borong Zhang, Ce Bian, Chao Yin, Chenxu Lv, Da Pan, Dian Wang, Dong Yan, Fan Yang, Fei Deng, Feng Wang, Feng Liu, Guangwei Ai, Guosheng Dong, Haizhou Zhao, Hang Xu, Haoze Sun, Hongda Zhang, Hui Liu, Jiaming Ji, Jian Xie, JunTao Dai, Kun Fang, Lei Su, Liang Song, Lifeng Liu, Liyun Ru, Luyao Ma, Mang Wang, Mickel Liu, MingAn Lin, Nuolan Nie, Peidong Guo, Ruiyang Sun, Tao Zhang, Tianpeng Li, Tianyu Li, Wei Cheng, Weipeng Chen, Xiangrong Zeng, Xiaochuan Wang, Xiaoxi Chen, Xin Men, Xin Yu, Xuehai Pan, Yanjun Shen, Yiding Wang, Yiyu Li, Youxin Jiang, Yuchen Gao, Yupeng Zhang, Zenan Zhou, and Zhiying Wu. Baichuan 2: Open large-scale language models, 2023.
- [44] Yizhi Li, Ruibin Yuan, Ge Zhang, Yinghao Ma, Xingran Chen, Hanzhi Yin, Chenghao Xiao, Chenghua Lin, Anton Ragni, Emmanouil Benetos, et al. Mert: Acoustic music understanding model with large-scale self-supervised training. *arXiv preprint arXiv:2306.00107*, 2023.
- [45] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- [46] Jiaqi Wang, Hanqi Jiang, Yiheng Liu, Chong Ma, Xu Zhang, Yi Pan, Mengyuan Liu, Peiran Gu, Sichen Xia, Wenjun Li, Yutong Zhang, Zihao Wu, Zhengliang Liu, Tianyang Zhong, Bao Ge, Tuo Zhang, Ning Qiang, Xintao Hu, Xi Jiang, Xin Zhang, Wei Zhang, Dinggang Shen, Tianming Liu, and Shu Zhang. A comprehensive review of multimodal large language models: Performance and challenges across different tasks, 2024.
- [47] James A Hanley and Barbara J McNeil. The meaning and use of the area under a receiver operating characteristic (roc) curve. *Radiology*, 143(1):29–36, 1982.
- [48] Herve Jégou, Matthijs Douze, and Cordelia Schmid. Product quantization for nearest neighbor search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(1):117–128, 2011.
- [49] Aaron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning. In *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS'17*, page 6309–6318, Red Hook, NY, USA, 2017. Curran Associates Inc.
- [50] Sohrab Ferdowsi, Slava Voloshynovskiy, and Dimche Kostadinov. Regularized residual quantization: a multi-layer sparse dictionary learning approach, 2017.
- [51] Maxim Naumov, Dheevatsa Mudigere, Hao-Jun Michael Shi, Jianyu Huang, Narayanan Sundaraman, Jongsoo Park, Xiaodong Wang, Udit Gupta, Carole-Jean Wu, Alisson G. Azzolini, Dmytro Dzhulgakov, Andrey Malleevich, Ilia Cherniavskii, Yinghai Lu, Raghuraman Krishnamoorthi, Ansha Yu, Volodymyr Kondratenko, Stephanie Pereira, Xianjie Chen, Wenlin Chen, Vijay Rao, Bill Jia, Liang Xiong, and Misha Smelyanskiy. Deep learning recommendation model for personalization and recommendation systems, 2019.